A General Example

Top Previous Next

What follows in this chapter is a Tutorial instruction for the design of the propeller for a 70m twin-screw Research Vessel. This example is used to illustrate many of the program's various pages and functions, and will be a good introduction to the operation of PropElements.

Establishing Requirements

Top Previous Next

This example is an extension of a tutorial for the ***HydroComp NavCad®*** software. PropElements is for detail propeller design, so you would typically use a program like NavCad for the analysis and development of system-level propeller design parameters before beginning a detail propeller design project. (See Requirements for Propeller Design for additional information.)

Tutorial example parameters

So, using the process and data described above for an example 70m Research Vessel (as if previously conducted in NavCad), we have concluded that the requirements for our PropElements propeller design will be:

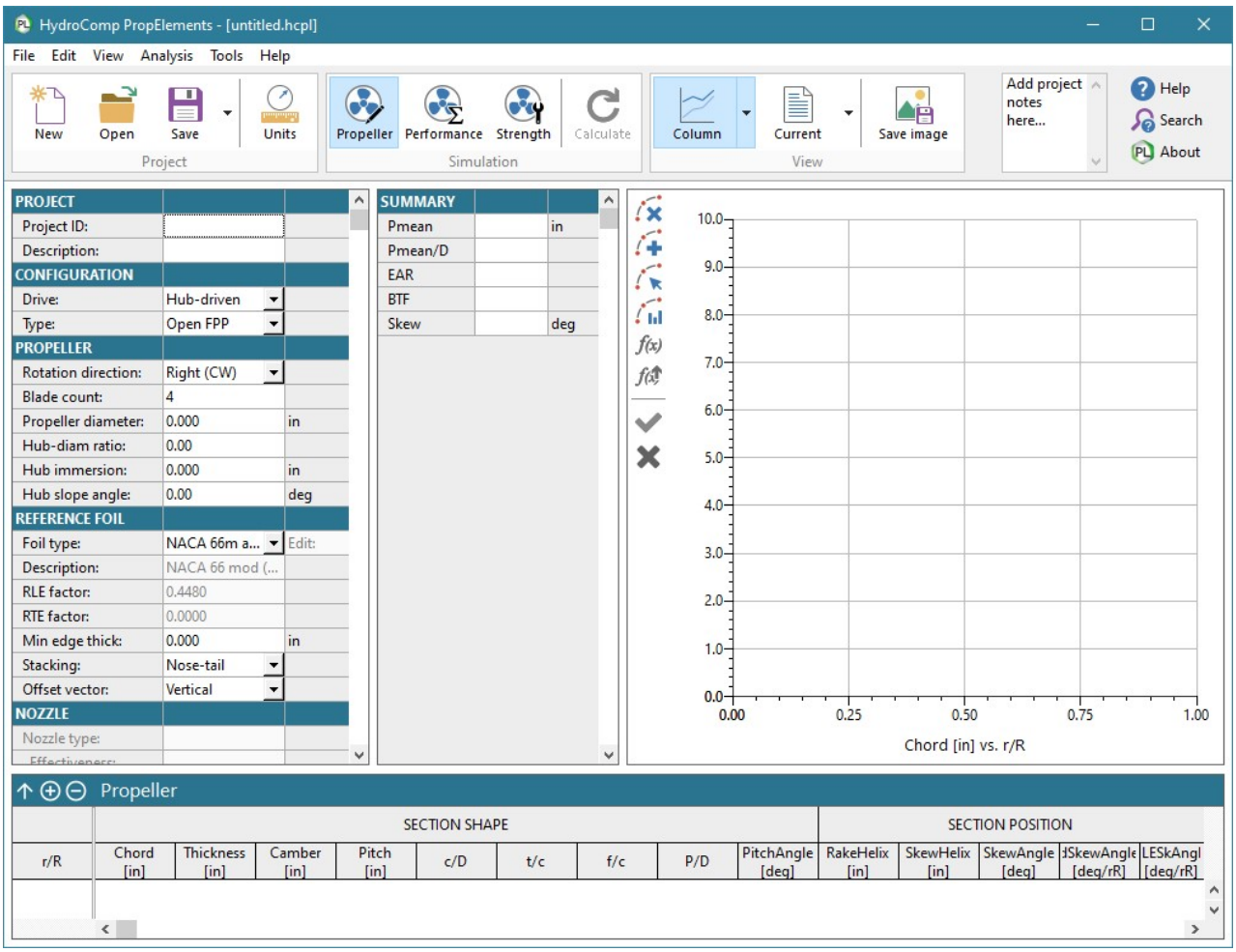
- style = **B Series outline**
- diameter = **2100 mm**
- number of blades = **4**
- expanded area ratio = **0.80**
- reference P/D ratio = **0.965**
- hub diameter = **Standard**
- vessel speed = **15.5 kts**
- wake fraction = **0.159**
- design shaft RPM = **375**
- design thrust = **217 kN** (open-water thrust per propeller)
- maximum developed power = **2750 kW**
- propeller immersion = **2720 mm**

As the designer, you will be responsible for defining much of the propeller. The following sections in this General Example will provide you with sample data and parameters.

Beginning a New Project

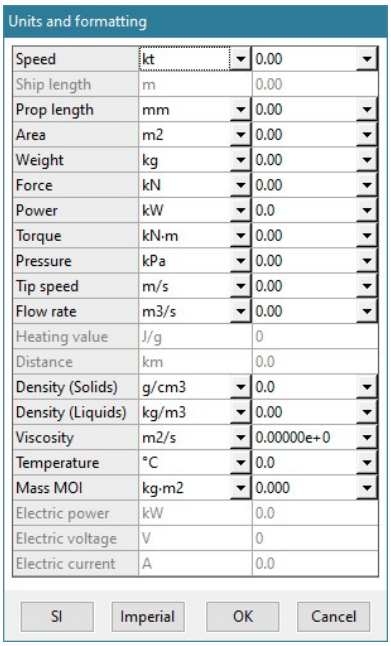
Top Previous Next

The *Propeller* page is the default starting page, and is where you will handle most of the file management duties, as well as entering general information about the radial parameters.



Setting project units

Set up your dimensional units and decimal formatting for the project.



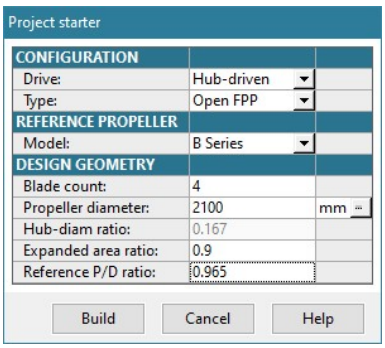
1. Click the menu **Tools | Units...** or the corresponding toolbar button shown here. You will note that only the units that are pertinent to PropElements are enabled.
2. Click on **SI** to set all of the units to SI (metric), then select the following specific units and formatting:
   Force = **kN [0.00]**
   Torque = **kN-m [0.00]**
3. Click **OK** to continue.

Beginning a new project

Follow these instructions to set up and enter data for your Tutorial design project.

1. To begin a clean project, click the menu **File | New project** or the corresponding toolbar button shown here. If you were working on an active design project, PropElements will check that you want to overwrite the active project and start new.

At this point, no data has been defined for the propeller. The first step is to define an initial design based on a few principal parameters.



1. Click **File | New...** from the main menu.
2. Enter or select the following data:
   Drive = **Hub-driven**
   Type = **Open FPP**
   Model = ***B Series***
   Blade count = **4**
   Propeller diameter = **2100 [mm]**
   Expanded area ratio = **0.9**
   Reference P/D ratio = **0.965**
3. Click **Build** to create the radial position distribution.

Enter new project data

To start defining your project, you will enter basic project identifier information and water properties into the parameters table, as well as project objectives and configuration.

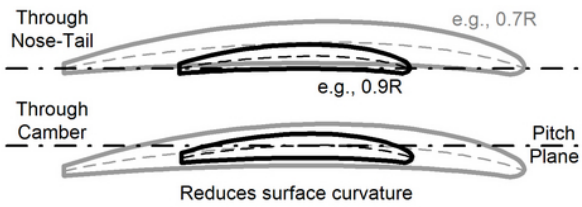
1. Enter the following basic *Project* group information:
   Project ID = **Tutorial**
   Description = **Example propeller design**
2. Continue by defining additional *Propeller* data for the project:
   Rotation direction = **Right (CW)**
   Hub immersion = **2720** mm
   Hub slope angle = **0**
3. Accept the default *Reference foil* parameters that were defined by the new project selection of the B Series model. The **NACA 16 a0.8** was pre-defined as the *Foil type (i.e., a NACA 16* thickness distribution with a=0.8 mean line camber). Foil shape information is currently used only for development of geometry for CAD export, and calculation of blade section properties for the *Strength* page calculation.) **Custom** foil geometries can also be defined using the foil editor by clicking the *Edit* button in the table. No correction for minimum edge thickness is needed. The default *Stacking* option is applied (*Nose-Tail*).
4. Finally, if a ducted propeller had been chosen (*Ducted FPP*), then additional selection options for the available nozzle styles will be enabled.

Blade foil stacking

One characteristic that is relevant to the blade strength is how the foil sections are "stacked". For example, the standard geometry definition places each foil section on its

design helix through the nose-tail line. However, for the purposes of smoothing surface curvature near the tip, the sections can be stacked on the helix along their maximum camber point, as shown in the graphic below. This will influence the calculation of the bending moment due to rake, and typically result in a total maximum stress that may be a few percent higher.

1. Select the following *Manufacture* setting:
   Foil stacking = select **None**



Save your project file

It is time to save your file with the data that you entered so far.

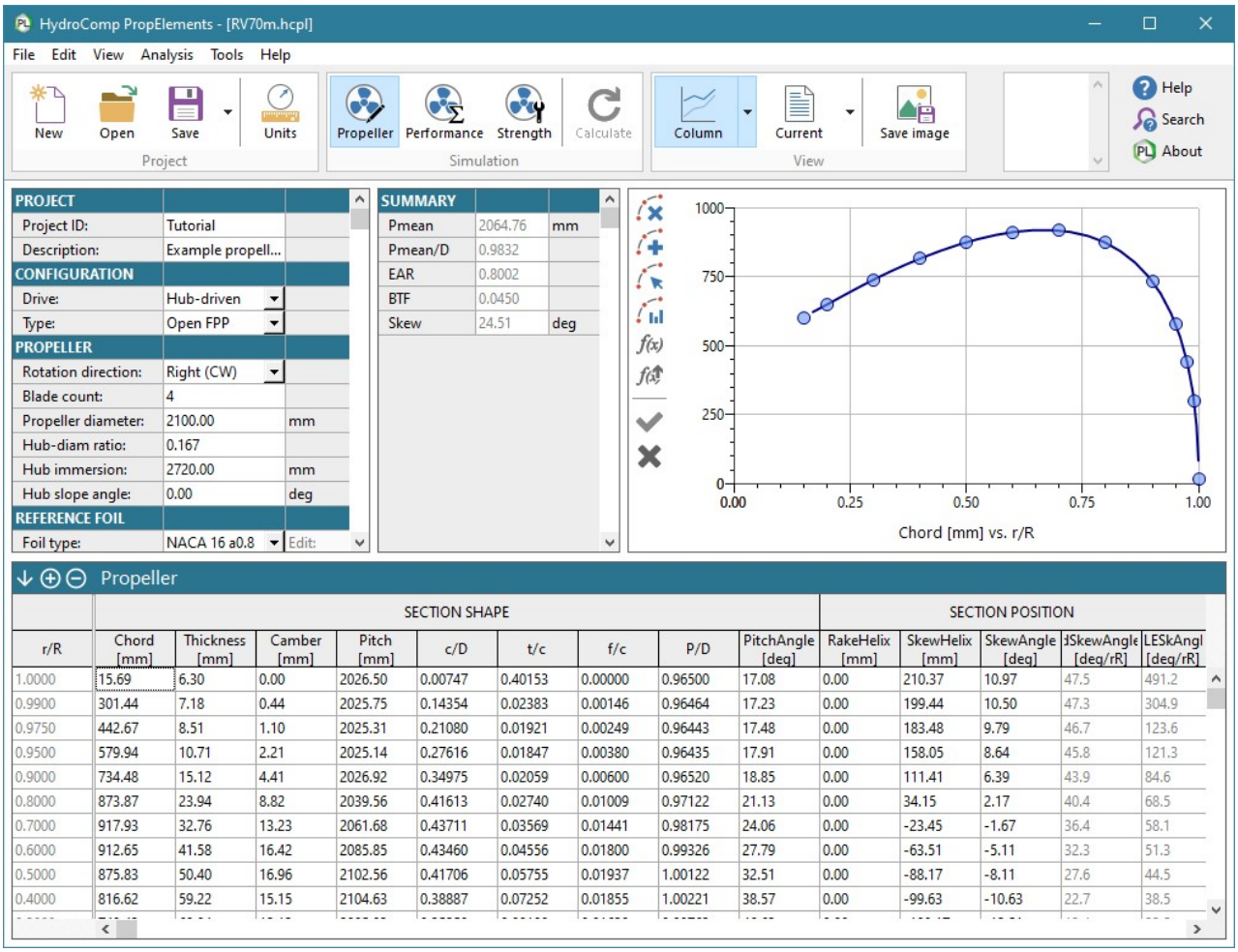
1. Click menu **File | Save project**, or just simply press *Ctrl+S*. Save the file to any directory on your computer with the file name **RV70m.hcpl**.

Note: The **HCPL** file extension (from **H**ydro**C**omp **P**ropE**l**ements) follows the open HydroComp Common Project File format. Information about this format can be found in the HydroComp Common Format Files help page.

Describing Radial Geometry

Top Previous Next

It is on the *Propeller* page that you will define and modify the principal radial geometric data. This data will include the *Pitch* and *Camber*, which can be optimized as part of the *Performance* page calculations, and it also includes data for *Chord*, *Thickness*, and *Skew angle* that will be manually reviewed and edited during the design process.



Enter initial radial data for our project

For this tutorial, we have used the New Project builder to describe a 4-bladed constant pitch propeller that is similar to a typical B Series.

NOTE: You may notice that the P/D ratio is somewhat different from the reference P/D ratio entered in the builder. It is important to remember that PropElements deals with nose-tail pitch reference, and not a traditional baseline reference such as used for the B Series.

If you need to manually enter radial data, such as from a propeller design drawing or table of parameters, you can copy and paste this information right into the columns.

Look at plots of the geometry

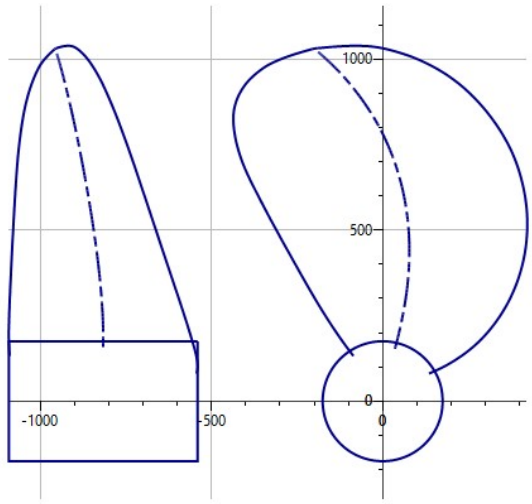
PropElements offers different plot options.

1. Click the menu **View | Column plot**, or the corresponding toolbar menu button shown here. (You can maximize your screen as desired.)

2. Click a column header (e.g., *Chord*) to put a column into the graph window. (Note: each page tracks its own selected column plots.)

Check the *2D layout* profile and outline plot

A traditional 2D drawing view of the propeller shape is with the *2D layout* profile and outline plot. This gives a sense of the propeller's shape as described by the radial distributions.

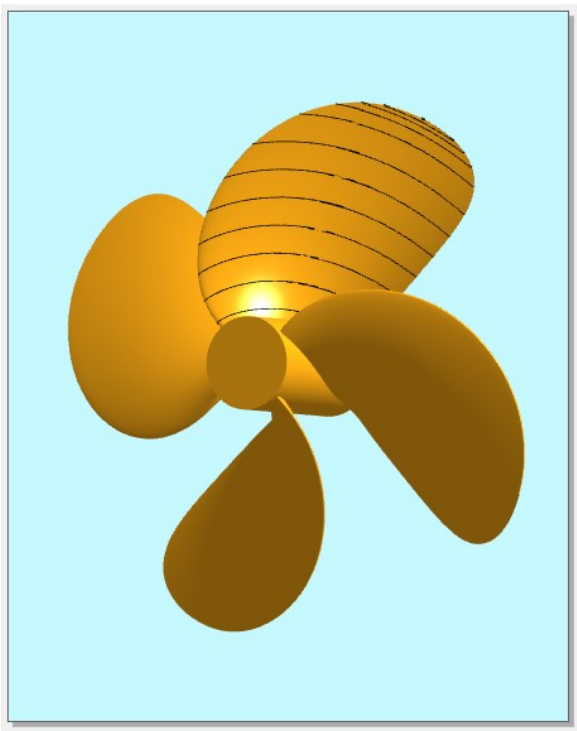


Profile and Outline

1. Click **View | 2D layout** from the main menu, or from the corresponding toolbar button menu.

Visualize with the *3D render* view

For a rendered image you can use the *3D render* view. This shows a rendered image of the propeller with representative section lines.



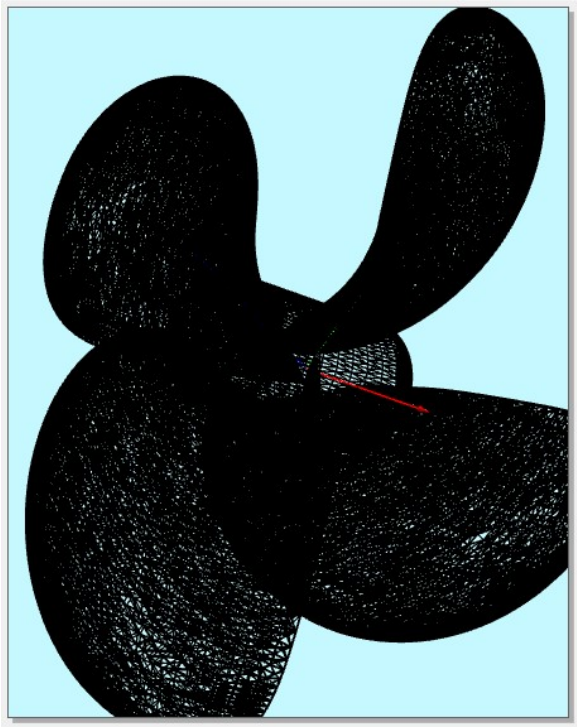
1. Click **View | 3D render** from the main menu, or from the corresponding toolbar button menu.
2. Save the view as BMP or IGES with the **Save plot** toolbar button.

You can manipulate the view by:

- Holding down left button and moving cursor for rotation
- Hold down right button and move cursor for in-out
- Also use mouse wheel for zoom in-out
- Re-click the menu item to refit and center the plot

Prepare a *3D mesh*

Convert the propeller into a *3D mesh*. The view displays the meshed objects.

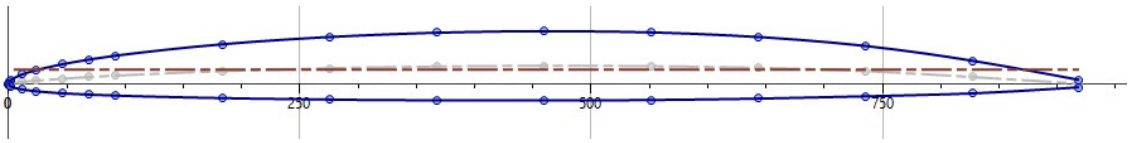


1. Click **View | Model | 3D mesh** from the main menu, or from the corresponding toolbar button menu.
2. Right-click the mesh view for a popup with the facet count.
3. Save the mesh as STL with the **Save plot** toolbar button.

You can manipulate the view in the same manner as the *3D render*.

See foil profiles for each section

Individual foil profiles can be viewed for each section. Each view is shown against the "nose-tail" line with the selected foil type, chord length and thickness, with the camber type and magnitude. Offset points are shown, as well as the neutral axis strength property. (Review Reference Foil Form for more information about how foil profiles are used in PropElements.)



1. You can expand the radial sections table by clicking the Expand toolbar button at the left of the table header.
2. Click any of the cells in the radius (r/R) column to view the associated foil profile.

Save your file

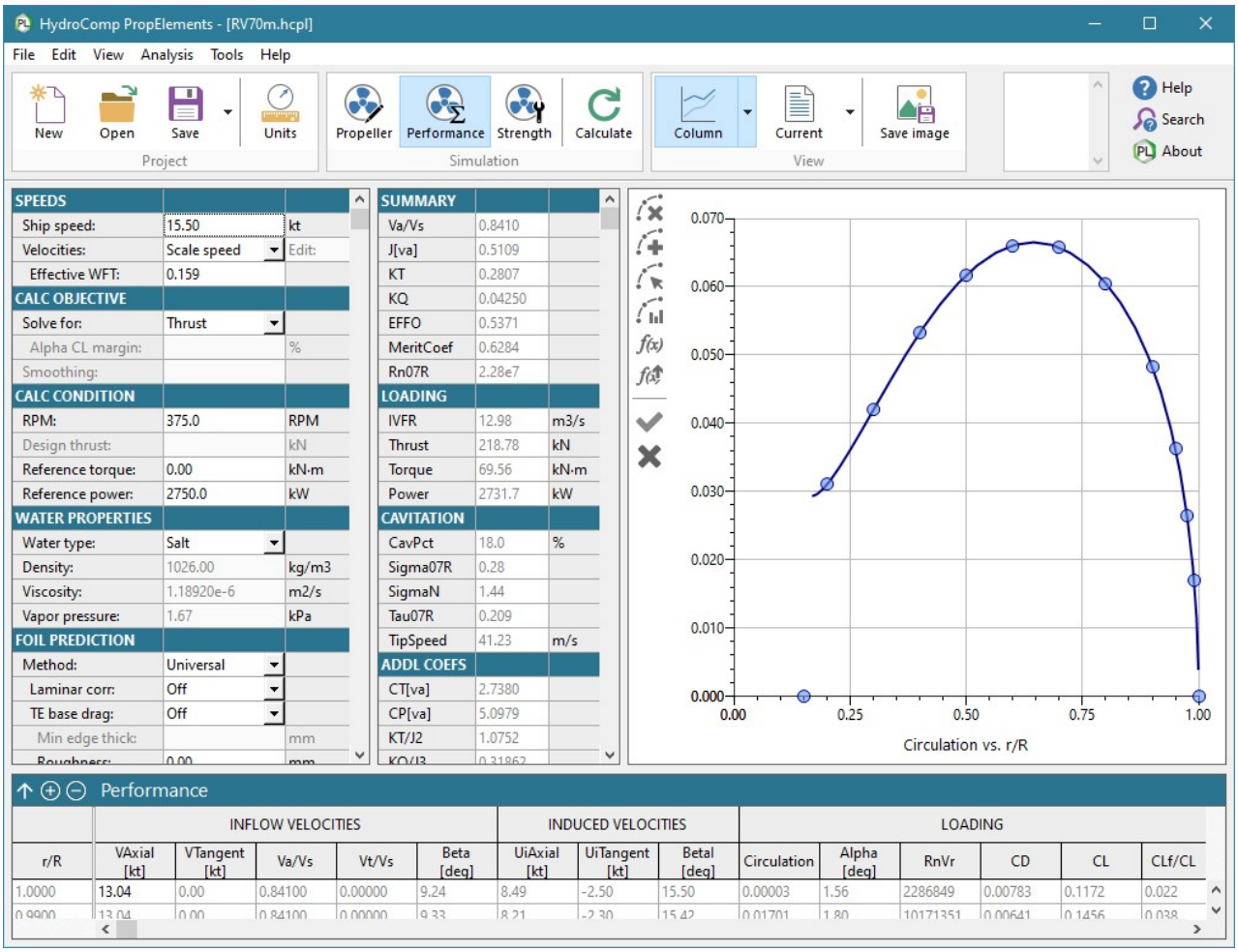
Periodically saving your file is always recommended.

1. Click menu **File | Save project**, the toolbar button, or press *Ctrl+S*. Your newly entered data will be saved to your **RV70m.hcpl** project file.

Analyzing Performance for a Known Geometry

Top Previous Next

The real work of the propeller design calculations occur on the *Performance* page. Here you will outline the objectives for the design calculations, as well as for the particular settings that reflect your technical preferences.



Enter design and analysis parameters

The geometry that was defined describes the principal shape and form parameters for the propeller. You will now define the performance objectives and analysis parameters.

1. Enter or select the following basic *Speeds* group information:
   Ship speed = **15.5** kts
   Velocities = **Scale from speed** (to scale the Speed to an entered wake fraction).
   Effective WFT = **0.159**
2. Continue by selecting the *Design objective* settings for the project:
   Solve for = **Thrust** (for an analysis of the given geometry; the corresponding pitch-camber design properties fields will be disabled)
3. The *Design condition* group establishes the operating mode and reference constraints:
   RPM = **375** (shaft RPM)
   Design thrust = since this is an analysis and not a design, this field is disabled
   Reference torque = omit this field for now
   Reference power = **2750** kW (the developed power that should not be exceeded)
4. Select or define the *Water properties* type for the project:
   Water type = **Salt** (the corresponding properties will be automatically set)
5. The lift and drag performance of each section is defined in this *Foil prediction* group:
   Method = **Universal** (the most general-case of the available options)
   Laminar corr = **Off** (this laminar/transition flow correction is typically only used for evaluation of model-scale propellers)
   TE base drag = **Off** (used for foils with a blunt trailing edge). The *Min edge thick* field is thereby disabled.
   Roughness = **0** mm (for no added roughness)
   Ideal angle (CL=1) = Click the estimate button and select **1.54** for the "a=0.8" meanline option. (This is a metric for the design inflow that is based on the character of the camber line, such as zero for a chord-wise symmetric camber; the field's estimate button will provide suggestions.)
6. Finally, the *Calculation* group is where the design objectives for radial loading of the propeller can be defined (such as to unload the tip). For this analysis, however, these design parameters are not required:
   Calc settings = **Standard**

Calculating performance

Click the menu **Analysis | Calculate: Performance** or the toolbar button to process the performance analysis. If the button is shown with the **red** option (as illustrated here), this indicates that some data has changed and a calculation update is needed.

Evaluating performance results

The overall performance of the propeller with the constant velocity distribution can be seen in the *Summary* table. A quick review tells us that:

1. The propeller meets the thrust requirement (of 217 kN), as well as the maximum delivered power limit (2750 kW).
2. The predicted cavitation percentage (using a "Burrill chart" analysis) is nominally 18%, which is a bit higher than recommended for commercial operation and which should be considered in greater detail if operation is required to be largely cavitation-free.
3. The propeller open water efficiency is approximately 0.54, which is of the expected magnitude for a propeller these parameters.

Reviewing performance at each radius

You will want to check the *CL* column for indications of excessive loading, as well as the *Sigma*, *ChordMin* and *RLEMin* columns for comparisons to local cavitation criteria.

CAVITATION		
Sigma	ChordMin [mm]	RLEMin [mm]
0.132	0.00	0.00
0.134	**352.54**	**4.55**
0.139	473.91	**6.04**
0.146	606.64	**7.78**
0.163	767.24	**9.84**
0.208	896.48	**11.65**
0.272	895.72	**10.49**
0.368	813.05	**9.42**
0.523	677.28	**9.48**
0.790	517.13	**10.01**
1.293	358.82	**9.78**
2.334	230.84	**6.84**
0.000	0.00	0.00

The cells highlighted in **red** indicate the incidence of cavitation is expected. (See Design Data Checks for additional details about the criteria used for these checks.) Fields in bold significantly exceed the criteria and should be considered for correction. In this example, we need additional chord length (greater blade width) at the outer radii. In later steps of this general example, we will modify the chord distribution, as well as introduce greater detail for the velocity field. As part of the optimization for this example, we will see how modifying the camber also changes the recommended minimum nose radius (RLEMin).

Save your file

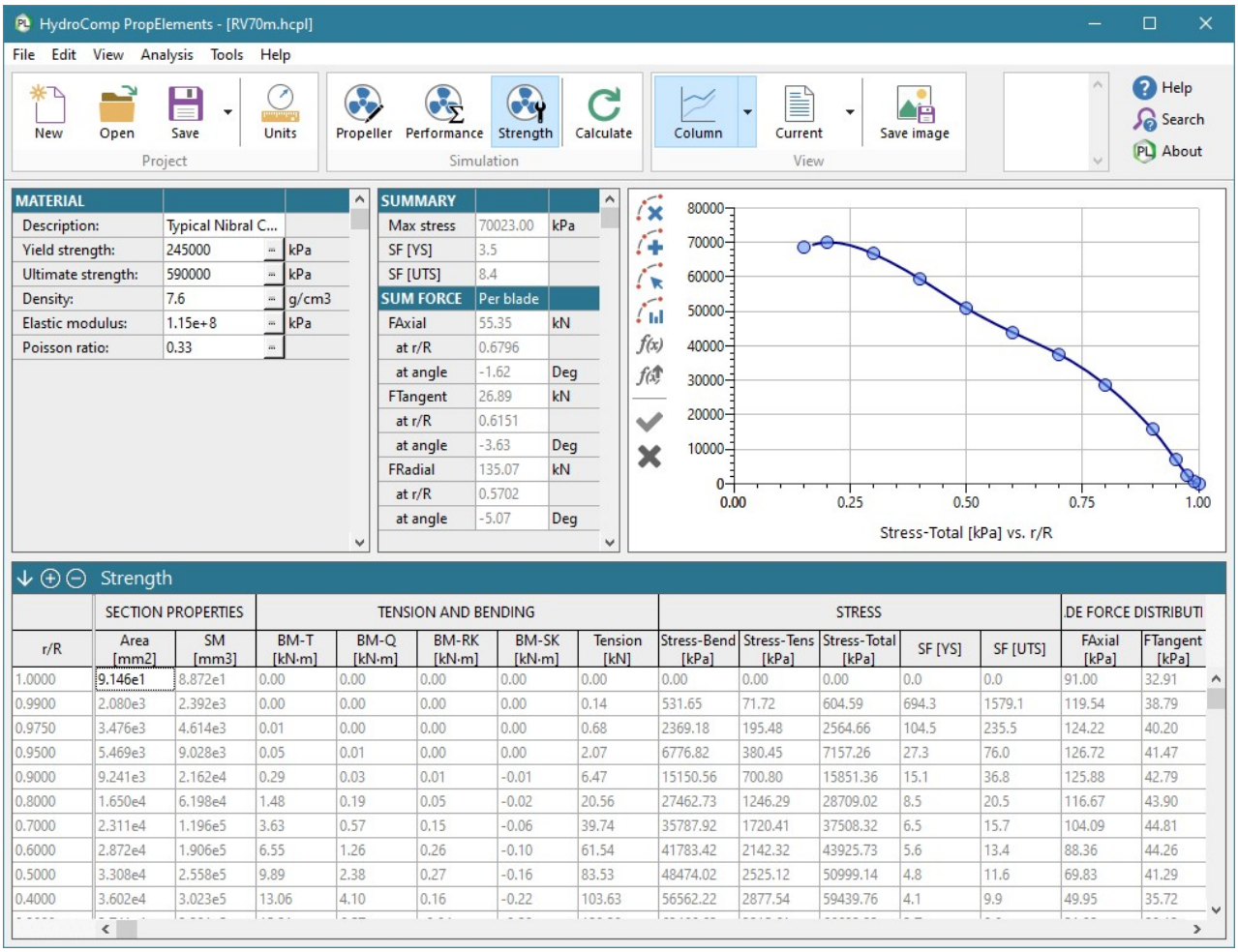
Periodically saving your file is always recommended.

1. Click menu **File | Save project**, or press *Ctrl+S*. Your newly entered data will be saved to your **RV70m.hcpl** project file.

Evaluating Blade Strength

Top Previous Next

The **Strength** analysis in PropElements is based on a cantilever beam calculation incorporating the effects of bending and tension due to thrust, torque, and centrifugal offset loading (due to skew and rake).



	SECTION PROPERTIES		TENSION AND BENDING					STRESS					DE FORCE DISTRIBUTI	
r/R	Area [mm2]	SM [mm3]	BM-T [kN·m]	BM-Q [kN·m]	BM-RK [kN·m]	BM-SK [kN·m]	Tension [kN]	Stress-Bend [kPa]	Stress-Tens [kPa]	Stress-Total [kPa]	SF [YS]	SF [UTS]	FAxial [kPa]	FTangent [kPa]
1.0000	9.146e1	8.872e1	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.0	0.0	91.00	32.91
0.9900	2.080e3	2.392e3	0.00	0.00	0.00	0.00	0.14	531.65	71.72	604.59	694.3	1579.1	119.54	38.79
0.9750	3.476e3	4.614e3	0.01	0.00	0.00	0.00	0.68	2369.18	195.48	2564.66	104.5	235.5	124.22	40.20
0.9500	5.469e3	9.028e3	0.05	0.01	0.00	0.00	2.07	6776.82	380.45	7157.26	27.3	76.0	126.72	41.47
0.9000	9.241e3	2.162e4	0.29	0.03	0.01	-0.01	6.47	15150.56	700.80	15851.36	15.1	36.8	125.88	42.79
0.8000	1.650e4	6.198e4	1.48	0.19	0.05	-0.02	20.56	27462.73	1246.29	28709.02	8.5	20.5	116.67	43.90
0.7000	2.311e4	1.196e5	3.63	0.57	0.15	-0.06	39.74	35787.92	1720.41	37508.32	6.5	15.7	104.09	44.81
0.6000	2.872e4	1.906e5	6.55	1.26	0.26	-0.10	61.54	41783.42	2142.32	43925.73	5.6	13.4	88.36	44.26
0.5000	3.308e4	2.558e5	9.89	2.38	0.27	-0.16	83.53	48474.02	2525.12	50999.14	4.8	11.6	69.83	41.29
0.4000	3.602e4	3.023e5	13.06	4.10	0.16	-0.22	103.63	56562.22	2877.54	59439.76	4.1	9.9	49.95	35.72

Strength calculation methodology

It is important to note that this is not a finite element analysis calculation. It is an adaptation of cantilever beam theory for the nature of propeller blades. It is for the global stress throughout the critical blade sections (typically at the root), and not for the prediction of a local stress concentration.

As a beam, it also therefore has limitations as to the amount of skew that can be used before the calculation is considered invalid. Testing and validation studies have shown good global stress prediction up to about 25 degrees skew.

The propeller's material properties

PropElements allows for the custom definition of the useful properties for the strength analysis, with estimated provided for typical marine propeller alloys (manganese bronze, nickel aluminum bronze, stainless steel).

1. Enter and select the following *Material* data:
   Description = enter **Typical Nibral**
   Yield strength = click the estimate button and select the value for **Ni Al bronze [CU3]**
   Ultimate strength = likewise, click and select the estimate for **Ni Al bronze [CU3]**
   Density = repeat for **Ni Al bronze [CU3]**
   Elastic modulus = repeat for **Ni Al bronze [CU3]**
   Poisson ratio = repeat for **Ni Al bronze [CU3]**

Calculating blade strength

Click the menu **Analysis | Calculate: Strength** or the toolbar button to process the blade strength analysis. If the button is shown with the **red** option (as illustrated here), this indicates that some data has changed and a calculation update is needed.

Evaluating strength results

Maximum values for the total stress (*Max stress*) and safety factors (*SF [YS]*, *SF [UTS]*) are determined from the calculated radial figures. Safety factors are the typical measures again the defined yield strength and ultimate (tensile) strength.

Determining an acceptable safety factor is often a matter of person preference, reflecting a subjective look at the role of the propeller (e.g., operation in ice versus clean flow), and even dynamic issues (e.g., potential for crash stop operation). Some references suggest safety factors against ultimate strength (*SF [UTS])* as high as 20 for extreme operating conditions, such as crash-back [Lee 2002].

Our project is not so extreme, so we can accept the design as having sufficient strength.

Force distribution and equivalent point forces

Within the *FORCE DISTRIBUTION* columns, you will note radial values for axial, tangential, and radial forces distributed per chord length and delta-radius. This then takes on its unit as pressure, but while this "pressure" is radially varying it is chordwise uniform. For example, if you wanted to utilize these figures for FEA, you can distribute the pressures evenly along the chord length. (This is an approximation for the chordwise pressure distribution, but one that is very useful for strength analysis - particularly for contemporary propeller blade foil sections with relatively flat pressure distributions.) The axial and tangential force distributions can also be used for CFD body forces.

In the *Summary* table, these force distribution have been integrated as single point forces and positions (by r/R and circumferential angle) for simplification with FEA computations.

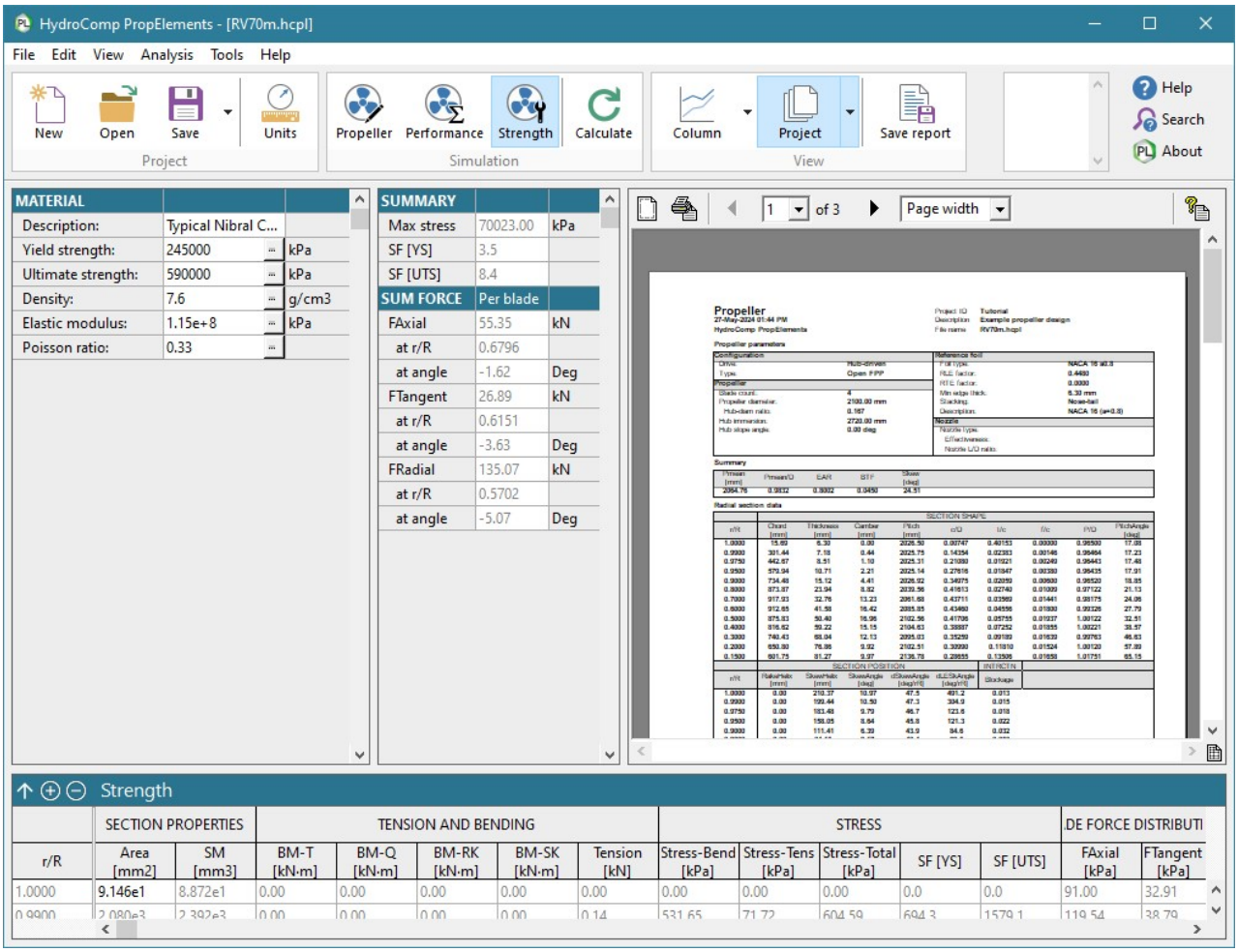
Save your file

Just a reminder!

Generating Reports

Top Previous Next

PropElements uses an integrated viewer to package the calculation data and results into a formatted report. Each page provides an opportunity to create report. These can be a single sheet report for the active PropElements page, or a multi-sheet report for the active page and all prior pages (e.g., the *Strength* page offers a three-sheet report including sheets for Propeller, Performance and Strength).



Assembling a multi-page report

Activated page groups will be placed into report pages for *Propeller*, *Performance*, and *Strength*.

1. Click the *Show all project pages* toolbar button (shown here) to generate a three-sheet report. Activated page groups will be placed into report pages for *Propeller*, *Performance*, and *Strength*.
2. Save the report to PDF (Adobe Acrobat), CSV (spreadsheet), or RTF (word processor) with the toolbar button.

Viewer toolbar

The report viewer provides viewing, printing and exporting of reports. Important features of the viewer are described in the toolbar sections shown below.

Document management

Page setup - setup page sizes, orientation, margins.

Print report - send the report to a printer.

Page navigation

Previous page - jump to the previous page.

1 of 1 *Page number* - jump to a page by the page number.

Next page - jump to the next page.

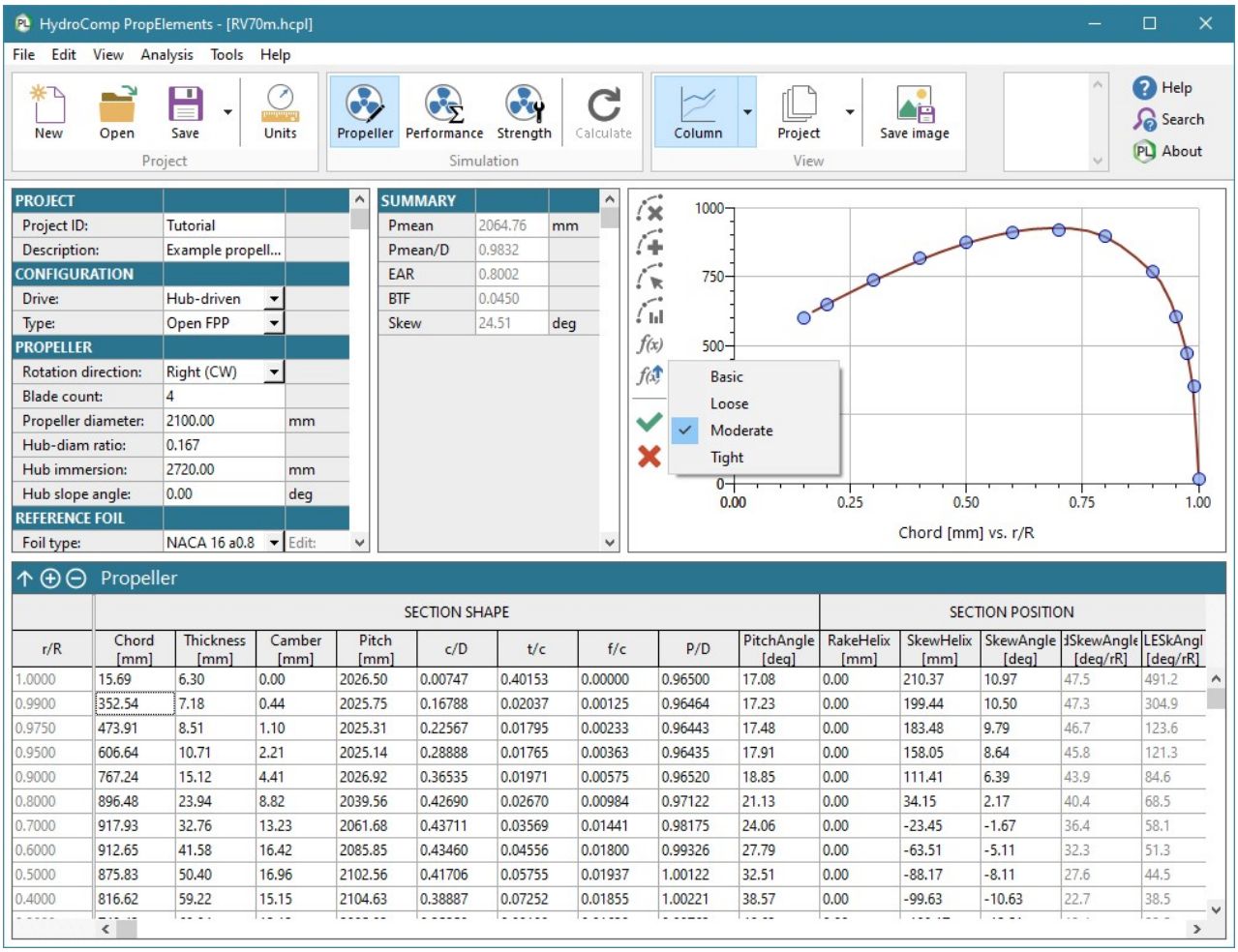
Zooming

100% *Zoom level* - zoom the page to Page width, Whole page, and various other zoom options. Zoom options are also available by right-clicking on the document.

Modifying Radial Geometry

Top Previous Next

PropElements offers a variety of ways to refine and modify the radial geometry. The previous analysis of performance for the initial geometry indicated that additional chord length at the outer radii are needed. This step of the example will describe how to paste a new column of *Chord* data and then to uniformly scale that column.



r/R	Chord [mm]	Thickness [mm]	Camber [mm]	Pitch [mm]	c/D	t/c	f/c	P/D	PitchAngle [deg]	RakeHelix [mm]	SkewHelix [mm]	SkewAngle [deg]	dSkewAngle [deg/rR]	LESkAngl [deg/rR]
1.0000	15.69	6.30	0.00	2026.50	0.00747	0.40153	0.00000	0.96500	17.08	0.00	210.37	10.97	47.5	491.2
0.9900	352.54	7.18	0.44	2025.75	0.16788	0.02037	0.00125	0.96464	17.23	0.00	199.44	10.50	47.3	304.9
0.9750	473.91	8.51	1.10	2025.31	0.22567	0.01795	0.00233	0.96443	17.48	0.00	183.48	9.79	46.7	123.6
0.9500	606.64	10.71	2.21	2025.14	0.28888	0.01765	0.00363	0.96435	17.91	0.00	158.05	8.64	45.8	121.3
0.9000	767.24	15.12	4.41	2026.92	0.36535	0.01971	0.00575	0.96520	18.85	0.00	111.41	6.39	43.9	84.6
0.8000	896.48	23.94	8.82	2039.56	0.42690	0.02670	0.00984	0.97122	21.13	0.00	34.15	2.17	40.4	68.5
0.7000	917.93	32.76	13.23	2061.68	0.43711	0.03569	0.01441	0.98175	24.06	0.00	-23.45	-1.67	36.4	58.1
0.6000	912.65	41.58	16.42	2085.85	0.43460	0.04556	0.01800	0.99326	27.79	0.00	-63.51	-5.11	32.3	51.3
0.5000	875.83	50.40	16.96	2102.56	0.41706	0.05755	0.01937	1.00122	32.51	0.00	-88.17	-8.11	27.6	44.5
0.4000	816.62	59.22	15.15	2104.63	0.38887	0.07252	0.01855	1.00221	38.57	0.00	-99.63	-10.63	22.7	38.5

Define a new chord distribution

As noted, this initial data for this tutorial was based on a standard B Series propeller. We will update the chord distribution for the indicated increases and then smooth the column.

1. Click back to the *Performance* page, then highlight and copy the cells that are colored red in *CAVITATION* group *ChordMin* column.
2. Click to the *Propeller* page and paste these into the proper cells in the **Chord** column. You will see the new data in the plot.
3. To further smooth this data, click the **Smooth column** toolbar button to the left of the plot.
4. This enables the smoothing control toolbar buttons below.
   - **Spline type**. Options for a *Normal* spline distribution, or a special *Tip weighted* spline that for shapes that have more curvature toward the tip radii, such as the *Chord* length.
   - **Fit order**. Options for the degree of the fit, from lowest order to highest: *Basic*, *Loose*, *Moderate*, *Tight*.

   Try the different options until you get a shape that is satisfactory. (The example shown above was developed with a *Tip weighted* spline with a *Moderate* fit.) You will see the proposed new column data shown as a solid line, while the original data is shown as the markers.
5. Then click the **Accept** button to save the modifications.

Scale the a column (Chord) by a uniform magnitude

The performance analysis suggests, however, that we probably should increase the chord lengths even more to achieve a larger blade area ratio and a reduction in the cavitation percentage. (This will also increase the strength safety factors a bit.) So, the next step is to widen and scale the entire column of chord values. For this tutorial example, we will propose to increase the chord length values by an additional 5%.

1. To scale a column, click into that column (**Chord**), and then click the **Scale column** toolbar button in the vertical bar to the left of the chord plot.
2. In the popup data form, enter **1.05** as the *column scale multiplier*. Click **OK**.
3. To accept this data click the **Accept** toolbar button. (To revert to the original data, click the **Cancel** button.)

There are additional tools to modify column data, including:

- **Shift column**. Click into a column and then click the toolbar button. Enter the amount to shift.
- **Fit point**. Click into a cell, then click the toolbar button.

Save the file

Once again, save your file.

1. Click menu **File | Save project**, or press *Ctrl+S*.

Using a Defined Radial Wake Distribution

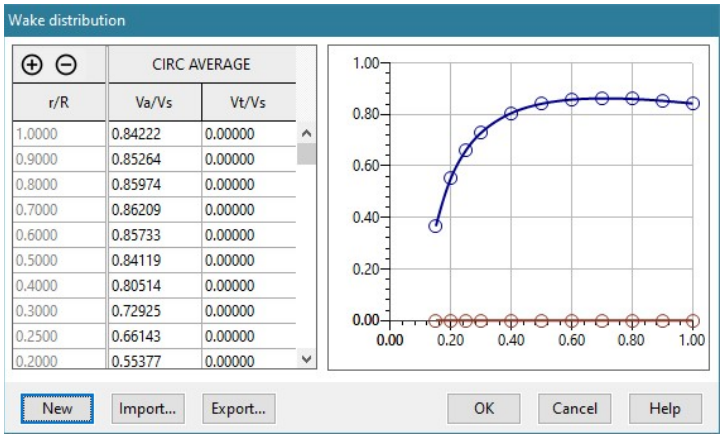
Top Previous Next

The initial study for this PropElements tutorial Example used a velocity distribution (*VAxial*) that was simply the ship speed scaled with the entered effective wake fraction. One of the significant technical values of a tool like PropElements is the ability to design the propeller's radial parameters so that they are suited to the ship's radial variation of the inflow velocity distribution. This distribution is called the "wake field" and it is employed for "wake-adapted" propeller design (as will be shown below). Note that this is the circumferential mean of the wake field velocities.

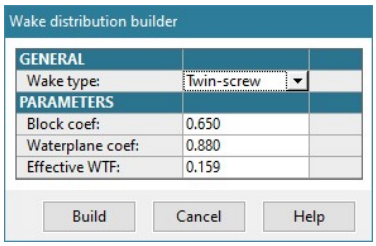
Describe a radial wake distribution

There are two parts to a radial wake distribution – the qualitative shape of the velocity distribution and the quantitative magnitude of the velocities. This is applied in PropElements by entering axial and tangential velocities that represent a "type-specific" distribution (or a "ship-specific" distribution if you have one), and scaling it to the entered effective wake fraction (*Effective WFT*). For this example, we will estimate a wake distribution using an algorithm that is representative of values for a twin-screw open-shaft vessel.

1. In the *Speeds* group on the *Performance* page, click and select:
   Velocities = **From wake dist** (to scale an entered value of wake fraction to a suitable radial distribution).
2. This enables the *Edit* button. Click this to display the *Wake distribution* page (as shown below).



3. Click **New**, then enter the following data for the 70m Research Vessel to estimate a representative open-shaft twin-screw mean velocity distribution. Once entered, defined click **Build**.



4. Click **OK** to accept this data.

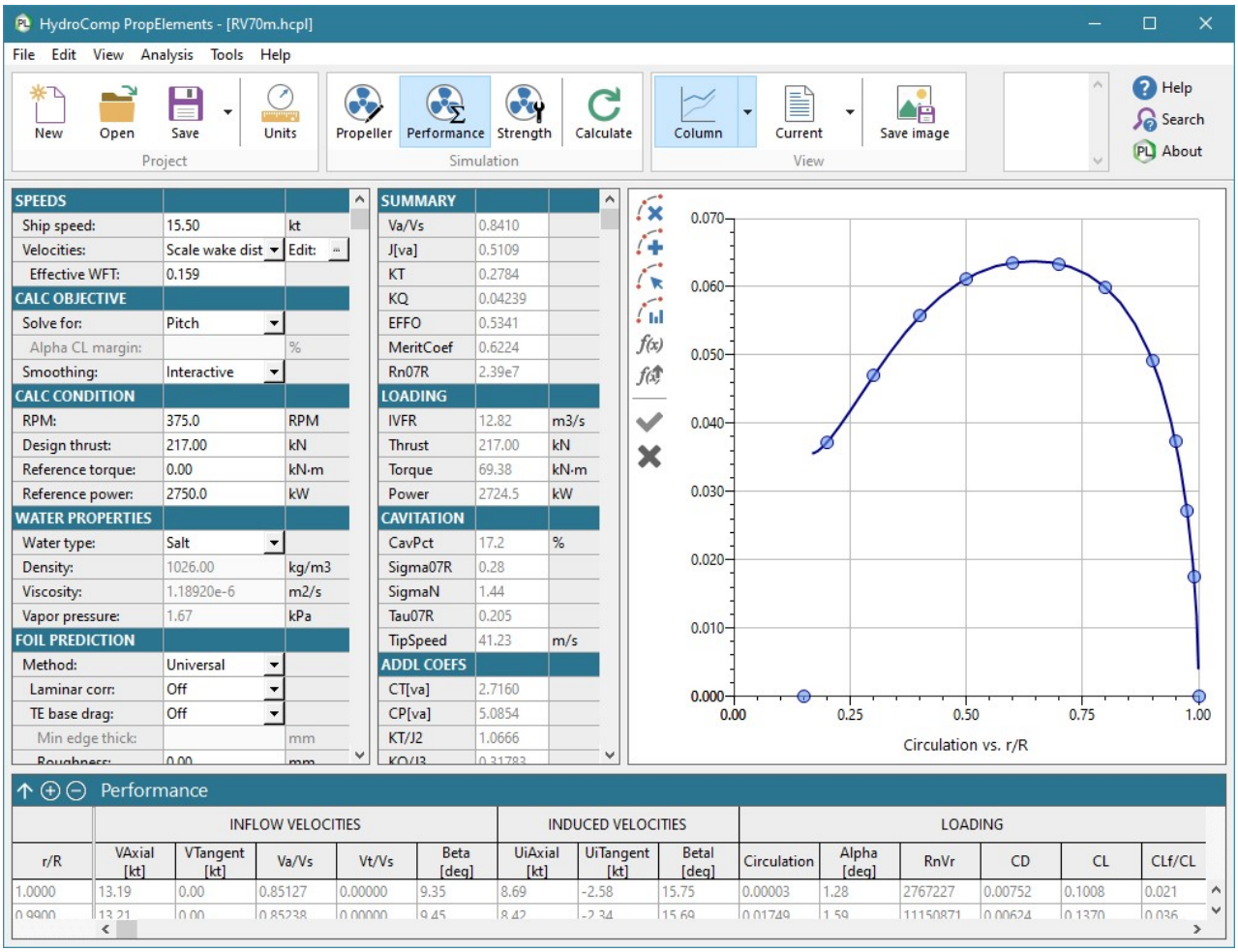
Note: Mean tangential velocities are typically small, so none are estimated in this prediction.

Designing for Optimum Performance

Top Previous Next

We have previously seen how PropElements evaluates performance for a known geometry. In other words, you have prepared a propeller analysis. The real work of wake-adapted propeller design is the next step in this tutorial Example. Here you will outline the objectives for the design calculations, as well as calculation settings that reflect your particular technical preferences.

The prior analysis confirmed that the propeller needed more chord length and that its operation would be in a velocity distribution (as defined by a representative wake file and effective wake fraction). We can now proceed to designing the pitch for optimum performance.



The design objectives and conditions

The purpose for this step is to design a wake-adapted pitch distribution for the described inflow velocities.

1. Select the following *Design objective* data:
   Solve for = **Pitch** (You can also design for *Pitch+Camber*, but this example illustrates a typical process used for the design of variable pitch only.)
   Smoothing = **Interactive** (This allows for interactive smoothing of the pitch distribution.)
2. Most of the *Design condition* data will have already been set, but we need to add the target thrust for the design:
   Ship speed = **15.5** kts
   RPM = **375** (Remember, this is the shaft RPM.)
   Design thrust = **217** kN
   Reference power = **2750** kW (the delivered power that should not be exceeded)
3. This example is using the **Standard** calculation settings for the defined *Hub-driven* drive and *Open FPP* propeller type. Additional *Custom* calculation settings include unloading the tip and/or hub, as well as reducing nozzle effectiveness (for the *Ducted FPP* type). See Designing for Noise Sensitive Operation for the use of custom settings.

Calculating performance

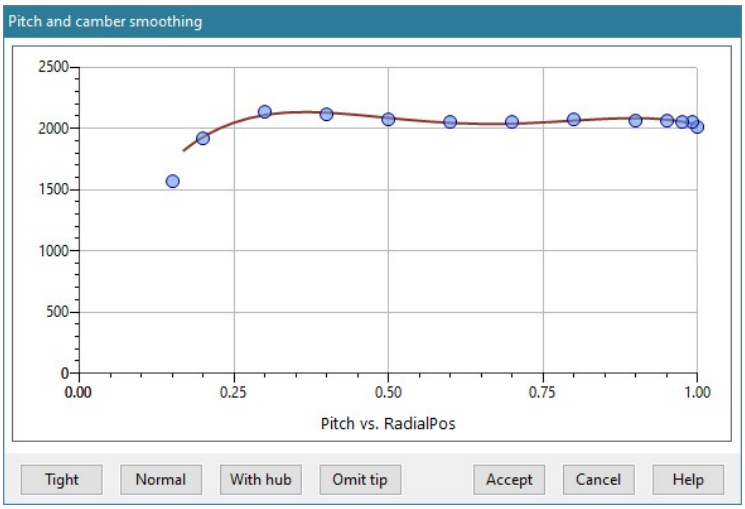
Click the **Calculate** page toolbar button to process the performance analysis. If the button is shown with the red option (shown here), this indicates that some data has changed, and a calculation update is needed.

Interactive smoothing

The numerical solution of the optimum characteristics of the propeller sometimes can result in unfair shapes. This is especially at the tip, where the combination of a very short chord (with a corresponding distorted thickness-chord and camber-chord ratios) can cause the design results to be "unfair".

It is therefore common to smooth the calculated design distributions to make them more friendly for follow-on "design for manufacture" (such as with the HydroComp PropCad software). The *Interactive* smoothing option allows you to smooth the optimized *Pitch* results (and *Camber*, as indicated) before it proceeds to a final analysis. A smoothing window is shown:

1. Click the tip control button to include (With tip) or exclude (Omit tip) the tip section from smoothing. Click until **Omit tip** is shown.
2. The center spline-type button can be used to change the type of fitting spline to provide more curvature near the tip. Leave this as **Normal**. Click the left-most fitting button until it shows **Tight**, for a reasonably smooth and tight fit.
3. Click the **Accept** button to proceed.



Evaluating performance results

The overall performance of the propeller can be seen in the *Summary* table. You will also want to check the *CL* column for indications of excessive loading, as well as the *Sigma*, *ChordMin* and *RLEMin* columns for comparisons to local cavitation criteria. (See Design Data Checks for additional information.)

Save your file

Again, please save your project.

1. Click menu **File | Save project**, or press *Ctrl+S*. Your performance data and results will be saved to your **RV70m.hcpl** project file.